

5 Communication Commands

5.1 Notes on Communication Commands

5.1.1 Command Suffixes

A command suffixed with **(X)** is an **X-firmware command**; suffixed with **(E)** is an **Emm-firmware command**.

No suffix or a **(Y42)** suffix indicates the command is **common to both firmware types**.

5.1.2 Partial Differences Between the Two Firmware Types

Differences in some commands between the two firmware types:

1. **Torque mode** — not available on Emm firmware; available on X firmware.
2. **Speed mode** — command format differs.
3. **Position mode** — command format differs; X firmware provides **two position modes**.
4. **Read system-status parameters** — command format differs.
5. **Modify system-status parameters** — command format differs.
6. **Read drive configuration parameters** — host TX format is the same, motor response format differs.
7. **Modify drive configuration parameters** — command format differs.
8. **Store a speed profile and auto-run on power-up** — command format differs.

Also, the **format for position angle / error** differs between the two firmware types. Except for the above, all other command formats are identical across the two firmware types.

5.1.3 Notes on Interrupting Position-Mode Commands

Regardless of X or Emm firmware, **Y42** supports **pre-emption** of a position-mode command: while a previous position command is still running, you can send a **new** position-mode command to change speed, acceleration, position, etc. The new command parameters are **updated in real time with a smooth transition**.

5.2 Trigger-Action Commands

5.2.1 Trigger Encoder Calibration

Host → Motor

Addr 06 45 6B

Motor → Host

Addr 06 02/E2/EE (see Ch.4) 6B

Example: send 01 06 45 6B → correct return 01 06 02 6B.

In **closed-loop** mode, this triggers encoder calibration: the motor slowly rotates one revolution CW, then one revolution CCW to linearize the encoder and improve accuracy. (*If not calibrated, error is $\pm 0.75^\circ \sim \pm 1.5^\circ$*). In **open-loop** mode, this returns error code E2.

5.2.2 Reboot Motor (Y42)

Host → Motor: Addr 08 97 6B

Motor → Host: Addr 08 02/E2/EE 6B

Example: 01 08 97 6B → 01 08 02 6B; the closed-loop motor resets and reboots.

Note: if a “power-down record” battery is connected, **position is retained** after reboot.

5.2.3 Clear Current Position Angle to Zero

Host → Motor: Addr 0A 6D 6B

Motor → Host: Addr 0A 02/E2/EE 6B

Example: 01 0A 6D 6B → 01 0A 02 6B; the **current position angle is zeroed**.

5.2.4 Clear Stall/Overheat/Overcurrent Protection

Host → Motor: Addr 0E 52 6B

Motor → Host: Addr 0E 02/E2/EE 6B

Example: 01 0E 52 6B → 01 0E 02 6B; clears stall, overheat, and overcurrent protection.

5.2.5 Restore Factory Settings

Host → Motor: Addr 0F 5F 6B

Motor → Host: Addr 0F 02/E2/EE 6B

Example: 01 0F 5F 6B → 01 0F 02 6B; restores factory settings. The green and blue LEDs flash alternately; **power-cycle**, then perform **no-load encoder calibration**.

5.3 Motion-Control Commands

5.3.1 Multi-Motor Command (Y42)

Host → Motor (broadcast frame)

Byte1 Byte2 Byte3 Byte4 Byte... ByteN

Addr FC Length Cmd#1 Cmd#2 ... Chk

00 AA 00xx 01 36 6B

or 02 36 6B

or 03 36 6B ...

6B

Motor → Host

Addr FC <data depends on sub-commands> 6B

Notes

1. For **all motion commands**, only **address 1** replies an **ack** to avoid bus collisions.
2. In multi-motor packets, a **position-mode** sub-command may include a **sync-flag** controlling “in-position reply”: send Addr + FC + 9F + 6B; 0 = no reply on reach; 1 = reply on reach.
3. If the multi-motor packet **contains read-parameter sub-commands**, avoid clashing with the **Addr=1 ack**. (*Ways: set Response to None/Reached, or change the motor’s address, etc.*)
4. **Modbus-RTU** does **not** support multi-motor packets.

Example (Emm firmware)

On the bus: motors at addresses **1, 2, 3, 4**. Requirements:

- **Motor 2**: speed 1500 RPM, accel 8, **relative** move -3600.0°, **no reply** on in-position.

- **Motor 3:** speed 1000 RPM, accel 10, **absolute** move 7200.0°, **reply** on in-position.
- **Motor 4: read** real-time position.

Send

00 AA 00 22

02 FD 01 05 DC 08 00 00 7D 00 00 00 6B

03 FD 00 03 E8 0A 00 00 FA 00 01 01 6B

04 36 6B

6B

Where:

- 00 broadcast; AA function code; 0022 = 34 bytes total; 6B checksum.
- 02 ... 6B = motor 2 position-mode move.
- 03 ... 6B = motor 3 position-mode move.
- 04 36 6B = motor 4 read real-time position (it will return position).
(Motor 2: no “in-position” reply; Motor 3: will reply when in position.)

5.3.2 Motor Enable Control

Host → Motor: Addr F3 AB <En=00/01> <Sync=00/01> 6B

Motor → Host: Addr F3 02/E2/EE 6B

- **Enable state:** 00/01 = disable/enable. (*Disabled = shaft de-energized/free to turn.*)
- **Sync flag:** 00/01 = execute immediately / buffer (see §3.5.2).

Example: 01 F3 AB 01 00 6B → 01 F3 02 6B; **enable** (lock the shaft).

5.3.3 Torque Mode Control (X)

Host → Motor: Addr F5 <Sign 00/01> <Slope 0000–FFFF mA/s> <Current 0000–1388 mA> <Sync 00/01> 6B

Motor → Host: Addr F5 02/9F/E2/EE 6B

(Parameters explained in §5.3.4.)

5.3.4 Torque Mode with Max-Speed Limit (X)

Host → Motor:

Addr C5 <Sign 00/01> <Slope 0000–FFFF mA/s> <Current 0000–1388 mA> <Sync 00/01> <MaxSpd 0000–7E30 (0–3000.0 RPM, 0.1 RPM units)> 6B

Motor → Host: Addr C5 02/9F/E2/EE 6B

- **Sign:** 00/01 = CW/CCW (i.e., positive/negative current).
- **Slope (accel):** 0–65535 mA/s.
- **Current:** 0–5000 mA.
- **Sync flag:** see §3.5.2.
- **Max speed:** 0–30000 (0.1 RPM → 0–3000.0 RPM).

Example: 01 C5 01 00 C8 02 58 00 0F A0 6B → 01 C5 02 6B

Meaning: CCW, slope 200 mA/s, current 600 mA, max speed 400.0 RPM.

Important: Torque mode exists **only** on X firmware. In torque mode the motor spins continuously per the **lower** of (commanded current capability vs. max speed). If the current setpoint is large, the motor will cap at the **max speed** during free run and will only reach the **current setpoint** when the gripper clamps (stalls). It then maintains that current/torque — suitable for **electric grippers, auto-winding**, etc.

5.3.5 Speed Mode (X)

Host → Motor: Addr F6 <Dir 00/01> <Accel 0000–FFFF RPM/s> <Speed 0000–7E30 (0.1 RPM)> <Sync 00/01> 6B

Motor → Host: Addr F6 02/E2/EE 6B

(Parameters explained in §5.3.6.)

5.3.6 Speed Mode with Max-Current Limit (X)

Host → Motor:

Addr C6 <Dir 00/01> <Accel 0000–FFFF RPM/s> <Speed 0000–7E30 (0.1 RPM)> <Sync 00/01> <Imax 0000–1388 mA> 6B

Motor → Host: Addr C6 02/E2/EE 6B

- **Dir:** 00/01 = CW/CCW
- **Accel:** 0–65535 RPM/s
- **Speed:** 0–30000 → 0–3000.0 RPM
- **Sync flag:** see §3.5.2

- **Max current:** 0–5000 mA

Example: 01 C6 01 03 E8 4E 20 00 07 D0 6B → 01 C6 02 6B
CCW, accel 1000 RPM/s, speed 2000.0 RPM, I_{max} 2000 mA.

5.3.7 Speed Mode (Emm)

Host → Motor: Addr F6 <Dir 00/01> <Speed 0000–0BB8 RPM> <Accel 00–FF (level)> <Sync 00/01> 6B

Motor → Host: Addr F6 02/E2/EE 6B

- **Dir:** 00/01 = CW/CCW
- **Speed:** 0–3000 RPM
- **Accel (levels 0–255):**
 - $\text{acc} = 0 \rightarrow$ **no S-curve**, start and run directly at set speed.
 - $\text{acc} \neq 0 \rightarrow$ S-curve time given by:
 $t_2 - t_1 = (256 - \text{acc}) * 50\mu\text{s}$, and $V_{t2} = V_{t1} + 1 \text{ RPM}$.
 Example: $\text{acc}=100 \rightarrow$ every $(256-100)*50\mu\text{s} = 7800\mu\text{s}$ speed changes by **1 RPM**.
 The larger acc, the **shorter** the accel time (faster ramp).
- **Sync flag:** see §3.5.2.

Example: 01 F6 01 05 DC 0A 00 6B → 01 F6 02 6B
CCW, 1500 RPM, accel level 10.

5.3.8 Pass-Through, Speed-Limited Position Mode (X)

Host → Motor:

Addr FB <Dir 00/01> <Speed 0000–7E30 (0.1 RPM)> <Pos 00000000–FFFFFFFF (0.1°)> <Mode 00/01/02> <Sync 00/01> 6B

Motor → Host: Addr FB 02/9F/E2/EE 6B
(Parameters explained in §5.3.9.)

5.3.9 Pass-Through, Speed-Limited Position Mode with I-Limit (X)

Host → Motor:

Addr CB <Dir 00/01> <Speed 0000–7530> <Pos 00000000–FFFFFFFF (0.1°)> <Mode 00/01/02> <Sync 00/01> <I_{max} 0000–1388 mA> 6B

Motor → Host: Addr CB 02/9F/E2/EE 6B

- **Dir:** 00/01 = CW/CCW

- **Speed:** 0–3000.0 RPM (0.1 RPM units)
- **Position angle:** 0–4294967295 (0.1° units; 1=0.1°, 10=1.0°, etc.)
- **Motion mode:**
 - 00 relative to **previous target** (incremental to last input target)
 - 01 absolute relative to **coordinate zero**
 - 02 relative to **current real-time position**
- **Sync flag:** see §3.5.2
- **I_{max}:** 0–5000 mA

Example: 01 CB 01 4E 20 00 00 8C A0 00 00 07 D0 6B → 01 CB 02 6B
 CCW, 2000.0 RPM, **relative** +3600.0°, I_{max} 2000 mA.

5.3.10 Trapezoidal Acc/Dec Position Mode (X)

Host → Motor:

Addr FD <Dir 00/01> <AccUp 0000–FFFF RPM/s> <AccDown 0000–FFFF RPM/s>
 <Vmax 0000–7530 (0.1 RPM)> <Pos 00000000–FFFFFFFF (0.1°)> <Mode 00/01/02>
 <Sync 00/01> 6B

Motor → Host: Addr FD 02/9F/E2/EE 6B
 (Parameters explained in §5.3.11.)

5.3.11 Trapezoidal Acc/Dec Position Mode with I-Limit (X)

Host → Motor:

Addr CD <Dir 00/01> <AccUp 0000–FFFF> <AccDown 0000–FFFF> <Vmax 0000–
 7530> <Pos 00000000–FFFFFFFF> <Mode 00/01/02> <Sync 00/01> <I_{max} 0000–
 1388> 6B

Motor → Host: Addr CD 02/9F/E2/EE 6B

- **Dir:** CW/CCW
- **AccUp/AccDown:** 0–65535 RPM/s
- **V_{max}:** 0–3000.0 RPM (0.1 RPM units)
- **Position angle:** 0–4294967295 (0.1° units)
- **Motion mode:** relative (prev target) / absolute (coord 0) / relative (current pos)

- **Sync flag:** see §3.5.2
- **Imax:** 0–5000 mA

Example:

01 CD 01 01 FF 01 FA 27 10 00 00 8C A0 00 00 07 D0 6B → 01 CD 02 6B

CCW, AccUp 511 RPM/s, AccDown 506 RPM/s, Vmax 1000.0 RPM, relative +3600.0°, Imax 2000 mA.

5.3.12 Position Mode (Emm)

Host → Motor:

Addr FD <Dir 00/01> <Speed 0000–0BB8 RPM> <Accel 00–FF lvl> <Pulses 00000000–FFFFFFFF> <Mode 00/01/02> <Sync 00/01> 6B

Motor → Host: Addr FD 02/9F/E2/EE 6B

- **Dir:** CW/CCW
- **Speed:** 0–3000 RPM
- **Accel level:** same rule as §5.3.7.
- **Pulses:** with default **1.8° motor, 1/16 microstep**, one pulse = $1.8^\circ/16 = 0.1125^\circ$; **3200 pulses = 360°**.
- **Mode:** relative (prev target) / absolute (coord 0) / relative (current pos)
- **Sync flag:** see §3.5.2

Example: 01 FD 01 05 DC 00 00 00 7D 00 00 00 6B → 01 FD 02 6B

CCW, 1500 RPM, accel level 0 (direct), relative **32000 pulses** (10 revs).

5.3.13 Immediate Stop

Host → Motor: Addr FE 98 <Sync 00/01> 6B

Motor → Host: Addr FE 02/E2/EE 6B

Sync flag: 00 immediate / 01 buffer (see §3.5.2).

Example: 01 FE 98 00 6B → 01 FE 02 6B; stops motor immediately.

5.3.14 Trigger Synchronous Motion (Buffered)

Host → Motor: Addr FF 66 6B

Motor → Host: Addr FF 02/E2/EE 6B

Example: 00 FF 66 6B (broadcast) → 01 FF 02 6B.

All motors that **have buffered commands** (i.e., previously sent with Sync=01) **start simultaneously**. Motors without a buffered command do nothing.

5.4 Homing Commands

5.4.1 Set Single-Turn Home Zero Position

Host → Motor: Addr 93 88 <Store 00/01> 6B

Motor → Host: Addr 93 02/E2/EE 6B

- **Store:** 00/01 = do not store / store (retained across power-down).

Example: 01 93 88 01 6B → 01 93 02 6B; sets **current position** as the **single-turn home zero** and **stores** it. *Note: this zero is within a **single revolution**.*

5.4.2 Trigger Homing

Host → Motor: Addr 9A <Mode 00–05> <Sync 00/01> 6B

Motor → Host: Addr 9A 02/9F/E2/EE 6B

- **Default Mode=00** (single-turn **nearest** home).
 - 00: single-turn nearest home
 - 01: single-turn directional home
 - 02: **limitless collision** homing
 - 03: **limit-switch** homing
 - 04: go to **absolute position coordinate zero**
 - 05: return to the **last power-down angle**

Sync flag: see §3.5.2.

Example: 01 9A 02 00 6B → 01 9A 02 6B; triggers **limitless collision homing**.

Important: For wiring/settings, see §3.2 “Homing to Origin” for each mode’s steps.

5.4.3 Force-Abort and Exit Homing

Host → Motor: Addr 9C 48 6B

Motor → Host: Addr 9C 02/E2/EE 6B

Example: 01 9C 48 6B → 01 9C 02 6B; abort homing in progress.

5.4.4 Read Homing Status Flags

Host → Motor: Addr 3B 6B

Motor → Host:

Addr 3B <Flags bit7..bit0> 6B

bit7 bit6 bit5 bit4 bit3 bit2 bit1 bit0

— — Ocp_TF Otp_TF Org_CF Org_SF Cal_Rdy Enc_Rdy

- **Ocp_TF:** over-current protection (0/1 = not triggered / triggered)
- **Otp_TF:** over-temperature protection (0/1 = not / triggered)
- **Org_CF:** homing **fail** flag
- **Org_SF:** homing **in progress** flag
 - Default: Org_CF=0, Org_SF=0
 - In progress: Org_CF=0, Org_SF=1
 - Failed: Org_CF=1, Org_SF=0
 - Success: Org_CF=0, Org_SF=0
- **Cal_Rdy:** calibration table ready (0/1 = not calibrated / calibrated)
- **Enc_Rdy:** encoder ready (0/1 = abnormal / normal)

Example: 01 3B 6B returns 01 3B 03 6B → flags 0x03 = 00000011b: Cal_Rdy=1, Enc_Rdy=1, others 0.

Code examples (testing individual bits):

```
if (flags & 0x01) Enc_Rdy = true; else Enc_Rdy = false;
```

```
if (flags & 0x02) Cal_Rdy = true; else Cal_Rdy = false;
```

```
if (flags & 0x04) Org_SF = true; else Org_SF = false;
```

```
if (flags & 0x08) Org_CF = true; else Org_CF = false;
```

```
if (flags & 0x10) Otp_TF = true; else Otp_TF = false;
```

```
if (flags & 0x20) Ocp_TF = true; else Ocp_TF = false;
```

Evaluate homing state:

```
int org_state = flags & 0x0C;
```

```
if (org_state == 0x04) { /* homing in progress */ }
```

```
if (org_state == 0x08) { /* homing failed */ }
```

```
if (org_state == 0x00) { /* homing success */ }
```

5.4.5 Read Homing Parameters

Host → Motor: Addr 22 6B

Motor → Host:

Addr 22 <Mode 00–05> <Dir 00/01> <HomeSpd 0000–0BB8 RPM> <Timeout
00000000–FFFFFFFF ms> <CollDetSpd 0000–0BB8 RPM> <CollDetCur 0000–1388
mA> <CollDetTime 0000–FFFF ms> <O_POT_En 00/01> 6B

(Parameters detailed in §5.4.6.)

5.4.6 Modify Homing Parameters

Host → Motor

Addr 4C AE <Store 00/01> <Mode 00–05> <Dir 00/01>

<HomeSpd 0000–0BB8 RPM>

<Timeout_Hi 0000–FFFF> <Timeout_Lo 0000–FFFF ms>

<CollDetSpd 0000–0BB8 RPM>

<CollDetCur 0000–1388 mA>

<CollDetTime 0000–FFFF ms>

<O_POT_En 00/01>

6B

Motor → Host: Addr 4C 02/E2/EE 6B

- **Mode** (default 00 = single-turn nearest):
00 nearest single-turn / 01 directional single-turn / 02 limitless collision / 03
limit-switch / 04 absolute coordinate zero / 05 last power-down angle
- **Dir:** 00/01 = CW/CCW (default CW)
- **Home speed:** 0–3000 RPM (default **30 RPM**)
- **Timeout:** in ms (default **10000 ms**); if exceeded, homing exits automatically.

- **Collision-homing detection thresholds** (defaults **300 RPM / 800 mA / 60 ms**):

Principle: Take the **stall point** as zero when all three hold true:

1. Real-time speed < CollDetSpd
2. Phase current > CollDetCur
3. Conditions (1)&(2) hold for > CollDetTime

- **O_POT_En**: power-on auto-homing enable (00/01 = disable/enable)

Example (write)

01 4C AE 01 00 00 00 1E 00 00 27 10 01 2C 03 20 00 3C 00 6B → 01 4C 02 6B

- 01 store; 00 Mode=Nearest; 00 Dir=CW;
- 001E HomeSpd=30 RPM; 00002710 Timeout=10000 ms;
- 012C CollDetSpd=300 RPM; 0320 CollDetCur=800 mA; 003C CollDetTime=60 ms;
- 00 O_POT_En = disabled.

Example (read)

01 22 6B → 01 22 00 00 00 1E 00 00 27 10 01 2C 03 20 00 3C 00 6B

- Mode = 00 (Nearest)
- Dir = 00 (CW)
- HomeSpd = 001E = 30 RPM
- Timeout = 00002710 = 10000 ms
- CollDetSpd = 012C = 300 RPM
- CollDetCur = 0320 = 800 mA
- CollDetTime = 003C = 60 ms
- Power-on auto-home = 00 (disabled)

5.5 Read System Parameter Commands

5.5.1 Timed Information Push (Y42)

Host → Motor

Addr 11 18 <InfoFC 36/3A/...> <Interval 0–FFFF ms> 6B

Motor → Host

Format follows the corresponding **Info Function Code** in §5.5.

Notes

1. Lets the motor **periodically push** info without frequent polling.
2. Supported function codes are those under **§5.5 Read System Parameter Commands**.

Examples

- Every **1 ms** push real-time position (FC=36): 01 11 18 36 00 01 6B; return format = §5.5.13.
- Every **10 ms** push status flags (FC=3A): 02 11 18 3A 00 0A 6B; return format = §5.5.15.
- **Stop** pushing status flags: 02 11 18 3A 00 00 6B; motor replies 02 11 6B.

5.5.2 Read FW/HW Version

Host → Motor: Addr 1F 6B

Motor → Host: Addr 1F <FW_Ver> <HW_Series> <HW_Type> <HW_Ver> 6B

- FW_Ver: e.g., 200 → **V2.0.0**
- HW_Series: 0/1 = X-series / Y-series board
- HW_Type: 0/1/2/3/4/5/6 = 20/28/35/42/57/86
- HW_Ver: e.g., 14 → **V1.4** (note in CN text 写“固件版本”为 14 但语义为硬件版本)

5.5.3 Read Phase Resistance & Inductance

Host → Motor: Addr 20 6B

Motor → Host: Addr 20 <Pha_R mΩ> <Pha_L uH> 6B

5.5.4 Read Bus Voltage

Host → Motor: Addr 24 6B

Motor → Host: Addr 24 <VBus mV> 6B

(V+ after reverse diode.)

5.5.5 Read Bus Current (Y42)

Host → Motor: Addr 26 6B

Motor → Host: Addr 26 <CBus mA> 6B

Note: Derived from phase current/voltage → **trend only**, may deviate.

5.5.6 Read Phase Current

Host → Motor: Addr 27 6B

Motor → Host: Addr 27 <CPha mA> 6B

5.5.7 Read Linearized Encoder Value

Host → Motor: Addr 31 6B

Motor → Host: Addr 31 <0–65535> 6B

Single-turn absolute: 0–65535 maps to **0–360°**, then wraps to 0.

5.5.8 Read Input Pulses

Host → Motor: Addr 32 6B

Motor → Host: Addr 32 <Sign 00/01> <Pulses 00000000–FFFFFFFF> 6B

- Sign: 00/01 = +/-
 - Pulses: at default **1.8° & 1/16, 3200 pulses = 360°**.
-

5.5.9 Read Target Position

Host → Motor: Addr 33 6B

Motor → Host: Addr 33 <Sign 00/01> <Target 00000000–FFFFFFFF> 6B

- **Emm:** 0–65535 = 0–360°. Angle = (Target×360)/65536.

- **X:** units **0.1°** → Angle = Target/10.
-

5.5.10 Read Real-Time Setpoint (Target Being Output)

Host → Motor: Addr 34 6B

Motor → Host: Addr 34 <Sign 00/01> <SetPos 00000000–FFFFFFFF> 6B

- **Emm:** angle = (SetPos×360)/65536.
 - **X:** angle = SetPos/10.
-

5.5.11 Read Real-Time Speed

Host → Motor: Addr 35 6B

Motor → Host: Addr 35 <0000–1388> 6B

- **Emm:** unit = RPM
 - **X:** unit = **0.1 RPM** → divide by 10.
-

5.5.12 Read Drive Temperature (Y42)

Host → Motor: Addr 39 6B

Motor → Host: Addr 39 <Sign 00/01> <Temp °C> 6B

Sign: 00/01 = negative/positive.

5.5.13 Read Real-Time Position

Host → Motor: Addr 36 6B

Motor → Host: Addr 36 <Sign 00/01> <Pos 00000000–FFFFFFFF> 6B

- **Emm:** angle = (Pos×360)/65536.
 - **X:** angle = **Pos/10** (0.1° units).
-

5.5.14 Read Position Error

Host → Motor: Addr 37 6B

Motor → Host: Addr 37 <Sign 00/01> <Err 00000000–FFFFFFFF> 6B

- **Emm:** angle = (Err×360)/65536.
- **X:** angle = **Err/100** (0.01° units).

5.5.15 Read Motor Status Flags

Host → Motor: Addr 3A 6B

Motor → Host: Addr 3A <Flags bit7..bit0> 6B

bit7 bit6 bit5 bit4 bit3 bit2 bit1 bit0

Oac — Esi_R Esi_L Cgp_TF Cgi_TF Prf_TF Ens_TF

- **Oac_TF:** power-down flag (defaults 1). You can write 0; after any power-loss it returns to 1.
- **Esi_R/L:** right/left limit switch input level (0=LOW, 1=HIGH).
- **Cgi/Cgp:** stall flag / stall-protection triggered.
Trigger protection when all hold:
 - (1) real-time speed < stall-speed threshold
 - (2) phase current > stall-current threshold
 - (3) (1)&(2) sustained > stall-time threshold
- **Prf_TF:** in-position (0/1 = not reached/reached).
- **Ens_TF:** enabled (0/1 = disabled/enabled).

Example: 01 3A 6B → 01 3A 83 6B ⇒ 0x83 = 10000011b → Oac_TF=1, Prf_TF=1, Ens_TF=1.

Bit-test examples (same 风格略) 见原文列表。

5.5.16 Read Homing Flags + Motor Status (Y42)

Host → Motor: Addr 3C 6B

Motor → Host: Addr 3C <HomeFlags> <StatusFlags> 6B

Interpretation per **§5.4.4** and **§5.5.15**.

5.5.17 Read IO Levels (Y42)

Host → Motor: Addr 3D 6B

Motor → Host: Addr 3D <bits> 6B

bit0 En_Pin (0=LOW, 1=HIGH)

bit2 Stp_Pin (0=LOW, 1=HIGH)

bit4 Dir_Pin (0=LOW, 1=HIGH)

bit5 Dir_OM (0=input, 1=output)

bit1/3/6/7 reserved

5.5.18 Read Battery Voltage (Y42)

Host → Motor: Addr 38 6B

Motor → Host: Addr 38 <VBat mV> 6B

(battery for low-power power-down record)

5.6 Drive Parameter R/W Commands

5.6.1 Modify Motor ID/Address

Host → Motor: Addr AE 4B <Store 00/01> <ID 01–FF> 6B

Motor → Host: Addr AE 02/E2/EE 6B

00 is broadcast; default ID=1.

5.6.2 Modify Microstep

Host → Motor: Addr 84 8A <Store 00/01> <Step 00–FF> 6B

Motor → Host: Addr 84 02/E2/EE 6B

01–FF=1–255; 00=256.

(At 1.8° & 1/16, Emm needs 3200 pulses per 360°.)

5.6.3 Modify Power-Down Flag

Host → Motor: Addr 50 <Oac 00/01> 6B → Addr 50 02/E2/EE 6B

Default 1; you may set 0; after power cycle it returns to **1**.

5.6.4 Read Option-State Parameters

Host → Motor: Addr 1A 6B

Motor → Host: Addr 1A <bits> 6B

bit0 MotType (0=1.8°, 1=0.9°)

bit1 FwType (0=X, 1=Emm)

bit2 CtrMode (0=open-loop, 1=closed-loop)

bit4 MotDir (0=CW, 1=CCW)

bit5 BtLock (0=keys unlocked, 1=locked)

bit7 Scale (0=×1, 1=angle input scaled ×0.1; i.e., 10× finer)

bit3/6 reserved

5.6.5 Modify Motor Type

Host → Motor: Addr D7 35 <Store 00/01> <Type 19/32> 6B

Motor → Host: Addr D7 02/E2/EE 6B

25/50 (原文写值=50→1.8°) 此处以你给定区间为准: **19/32 = 1.8°/0.9°** (保持原文语义: 据实设置后需空载重校准)。

5.6.6 Modify Firmware Type

Host → Motor: Addr D5 69 <Store 00/01> <Fw 00/01> 6B

Motor → Host: Addr D5 02/E2/EE 6B

00/01 = X/Emm. Modify when motor is stopped if possible.

5.6.7 Modify Control Mode (Open/Closed)

Host → Motor: Addr 46 69 <Store 00/01> <Mode 00/01> 6B

Motor → Host: Addr 46 02/E2/EE 6B

00/01 = open-loop/closed-loop (FOC). Prefer modifying at stop.

5.6.8 Modify Positive Motion Direction

Host → Motor: Addr D4 60 <Store 00/01> <Dir 00/01> 6B

Motor → Host: Addr D4 02/E2/EE 6B

00/01 = CW/CCW. (Also affected by swapping A+/A- or B+/B- then re-calibrating.)

5.6.9 Lock/Unlock Keys

Host → Motor: Addr D0 B3 <Store 00/01> <Lock 00/01> 6B

Motor → Host: Addr D0 02/E2/EE 6B

00/01 = unlock/lock. When locked, keys are disabled **except** factory reset; must unlock via command to re-enable.

5.6.10 Angle Command Extra ×0.1 Scaling (X)

Host → Motor: Addr 4F 71 <Store 00/01> <Enable 00/01> 6B

Motor → Host: Addr 4F 02/E2/EE 6B

Default 00. When enabled, the **angle input** becomes **0.01°** resolution (10× finer).

5.6.11 Speed Command ×0.1 Scaling (Emm)

Host → Motor: Addr 4F 71 <Store 00/01> <Enable 00/01> 6B

Motor → Host: Addr 4F 02/E2/EE 6B

Default 00. When enabled, **speed input** is in **0.1 RPM** (10× finer).

5.6.12 Open-Loop Working Current

Host → Motor: Addr 44 33 <Store 00/01> <I 0000–1388 mA> 6B

Motor → Host: Addr 44 02/E2/EE 6B

Default Y42/**1200 mA**. Avoid setting too high to reduce heating.

5.6.13 Closed-Loop Max Current

Host → Motor: Addr 45 66 <Store 00/01> <Imax 0000–1388 mA> 6B

Motor → Host: Addr 45 02/E2/EE 6B

Default Y42/**3000 mA**.

- **Emm:** stall **max** current cap.

- **X**: global max regardless of state.
-

5.6.14 Read PID (X)

Host → Motor: Addr 21 6B

Motor → Host: Addr 21 <pTkp> <pBkp> <vkp> <vki> 6B

(pTkp = trapezoid position Kp; pBkp = pass-through pos Kp; vkp/vki = speed loop gains.)

5.6.15 Modify PID (X)

Host → Motor:

Addr 4A C3 <Store 00/01> <pTkp> <pBkp> <vkp> <vki> 6B

Motor → Host: Addr 4A 02/E2/EE 6B

Defaults:

- pTkp/pBkp = (Y28/Y35/Y42/Y57)/**126640**
 - vkp = (Y28/Y35)/12600, **Y42/15600**, Y57/26000
 - vki = (Y28/Y35)/18, **Y42/26**, Y57/58
-

5.6.16 Read PID (Emm)

Host → Motor: Addr 21 6B

Motor → Host: Addr 21 <Kp> <Ki> <Kd> 6B

5.6.17 Modify PID (Emm)

Host → Motor: Addr 4A C3 <Store 00/01> <Kp> <Ki> <Kd> 6B

Motor → Host: Addr 4A 02/E2/EE 6B

Defaults: Kp=(Y28/Y35/Y42)/18000, Y57/32000; Ki=10; Kd 同 Kp。

5.6.18 Read DMX512 Parameters (Y42)

Host → Motor: Addr 49 78 6B

Motor → Host: Addr 49 <TotalCh 0001–0040> <ChPerMotor 01/02> <Mode 00/01>
<SingleChSpeed 0001–0BB8 RPM> <Accel 0001–FFFF> <DualChSpeedStep 0001–FFFF>
<DualChMoveStep 00000001–FFFFFFFF> 6B
(Details see next.)

5.6.19 Modify DMX512 Parameters (Y42)

Host → Motor:

Addr D9 90 <Store 00/01> <TotalCh 0001–0040> <ChPerMotor 01/02> <Mode 00/01>
<SingleChSpeed 0001–0BB8> <Accel 0001–FFFF> <DualChSpeedStep 0001–FFFF>
<DualChMoveStep 00000001–FFFFFFFF> 6B

Motor → Host: Addr D9 02/E2/EE 6B

- **TotalCh:** total channels of console (default **192**); must match controller.
 - **ChPerMotor:** 1 = single-channel (channel→position angle), 2 = dual-channel (**low**=position, **high**=speed).
 - **Mode:** 1 absolute coordinate position; 0 relative (incremental).
 - **SingleChSpeed:** default **1000 RPM**.
 - **Accel:** default **1000**; Emm: $\text{acc} = \text{value}/8$, X: **RPM/s**.
 - **DualChSpeedStep:** default **10** → $\text{speed} = (\text{channel value} \times 10) \text{ RPM}$.
 - **DualChMoveStep:** default **100** → $\text{angle} = (\text{channel value} \times 10.0)^\circ$.
-

5.6.20 Read In-Position Window (Y42)

Host → Motor: Addr 41 6B

Motor → Host: Addr 41 <Pos_Window 0000–FFFF> 6B
(Details see next.)

5.6.21 Modify In-Position Window (Y42)

Host → Motor: Addr D1 07 <Store 00/01> <Pos_Window 0000–FFFF> 6B

Motor → Host: Addr D1 02/E2/EE 6B

Default **8** (internally $\times 0.1$) $\rightarrow$ **0.8°**.

If $|\text{Target} - \text{Actual}| < \text{Window}$, set **Prf_TF**. With Response set to Reached/Both/Other, it returns Addr + FC + 9F + 6B.

5.6.22 Read Overheat/Overcurrent Thresholds (Y42)

Host $\rightarrow$ Motor: Addr 13 6B

Motor $\rightarrow$ Host: Addr 13 <OTP_Th 0000–FFFF> <OCP_Th 0000–FFFF> <OT/OC_Time 0000–FFFF ms> 6B

5.6.23 Modify Overheat/Overcurrent Thresholds (Y42)

Host $\rightarrow$ Motor: Addr D3 56 <Store 00/01> <OTP_Th> <OCP_Th> <OT/OC_Time> 6B

Motor $\rightarrow$ Host: Addr D3 02/E2/EE 6B

Defaults: 100°C / 6600 mA / 1000 ms. Protection triggers when **Temp/Current $\geq$ Th** for **$\geq$ Time**.

5.6.24 Read Heartbeat Timeout (Y42)

Host $\rightarrow$ Motor: Addr 16 6B

Motor $\rightarrow$ Host: Addr 16 <Timeout 00000000–FFFFFFFF ms> 6B

5.6.25 Modify Heartbeat Timeout (Y42)

Host $\rightarrow$ Motor: Addr 68 38 <Store 00/01> <Timeout 00000000–FFFFFFFF ms> 6B

Motor $\rightarrow$ Host: Addr 68 02/E2/EE 6B

Default 0. If no **valid** command within timeout, the drive **E-stops** to mitigate risk in realtime control when comms fail.

5.6.26 Read Integral Limit / Stiffness (X)

Host $\rightarrow$ Motor: Addr 23 6B

Motor $\rightarrow$ Host: Addr 23 <Value 00000000–FFFFFFFF> 6B

5.6.27 Modify Integral Limit / Stiffness (X)

Host → Motor: Addr 4B 57 <Store 00/01> <Value> 6B

Motor → Host: Addr 4B 02/E2/EE 6B

- **Emm:** integral limit default **65535**.
 - **X:** stiffness default Y42/**388**, Y57/**512**. Larger → higher static torque/stiffness, too large may cause oscillation.
-

5.7 Power-On Auto-Run

5.7.1 Store Speed Profile, Auto-Run on Power-Up (X)

Host → Motor:

Addr F7 1C <Clr/Store 00/01> <Dir 00/01> <Accel 0000–FFFF RPM/s> <Speed 0000–7530 (0.1 RPM)> <EnPinCtl 00/01> 6B

Motor → Host: Addr F7 02/E2/EE 6B

- 00/01 = clear / store.
 - Dir = CW/CCW.
 - EnPinCtl: enable **En-pin start/stop** (H/L/Hold configurable elsewhere).
-

5.7.2 Store Speed Profile, Auto-Run on Power-Up (Emm)

Host → Motor:

Addr F7 1C <Clr/Store 00/01> <Dir 00/01> <Speed 0000–0BB8 RPM> <Accel 00–FF lvl> <EnPinCtl 00/01> 6B

Motor → Host: Addr F7 02/E2/EE 6B

(Accel levels per §5.3.7). Example (600 RPM、acc=100、En-pin)。

5.8 Read All Drive Parameters

5.8.1 Read System Status (X)

Host → Motor: Addr 43 7A 6B

Motor → Host:

Addr 43 <Len=0x25> <Cnt=0x0C> <VBus mV> <CBus mA> <CPha mA> <EncRaw>

<EncLin> <S1 sign Target 0.1°> <S2 sign Speed 0.1RPM> <S3 sign Pos 0.1°> <S4 sign Err 0.01°> <S5 sign Temp °C> <HomeFlags> <StatusFlags> 6B

5.8.2 Read System Status (Emm)

Host → Motor: Addr 43 7A 6B

Motor → Host:

Addr 43 <Len=0x1F> <Cnt=0x09> <VBus mV> <CPha mA> <EncLin> <S1 sign Target 0–65535→angle> <S2 sign Speed RPM> <S3 sign Pos 0–65535→angle> <S4 sign Err 0–65535→angle> <HomeFlags> <StatusFlags> 6B

5.8.3 Read Drive Config (X)

Host → Motor: Addr 42 6C 6B

Motor → Host: returns **0x25 bytes, 0x18 params** including: key-lock, control-mode, pulse-port mux, comm-port mux, En/Dir polarity, microstep & interpolation, open/closed currents, closed-loop max speed, current-loop bandwidth, UART baud (00–08 → 9600...921600), CAN speed (00–09 → 10K...1M), **checksum mode** (00 fixed 6B / 01 XOR / 02 CRC-8 / 03 Modbus-RTU / 04 dual: free(6B)+DMX512), **response policy** (None/Receive/Reached/Both/Other), 0.1° scaling, stall-protection (enable, speed/current/time), in-position window (0.8° default).

5.8.4 Modify Drive Config (X)

Host → Motor: Addr 48 D1 <Store> <... 24 params ...> 6B

Motor → Host: Addr 48 02/E2/EE 6B

5.8.5 Read Drive Config (Emm)

Host → Motor: Addr 42 6C 6B

Motor → Host: **0x21 bytes, 0x15 params**: motor type, port mux, En/Dir polarity, microstep & interpolation, open current, **stall max current**, **max output voltage** (0000–1388 → (0–5000)×4 mV), baud/CAN, **ID**, checksum mode, response policy,

stall-protection thresholds, in-position window.

5.8.6 Modify Drive Config (Emm)

Host → Motor: Addr 48 D1 <Store> <... 21 params ...> 6B

Motor → Host: Addr 48 02/E2/EE 6B

- Pulse-port mux: OFF/OPEN/FOC/ESI_RCO/pLR_ESI
 - Comm-port mux: OFF/ESI_ALO/UART/CAN/uLR_ESI
 - Max output voltage: default 4000 → **4000×3 = 12000 mV** (per example)
 - DMX512: baud fixed 250000, stop bits = 2
-

6 FAQs and Notes

6.1 FAQs

See §1.6 LED Status Indications.

6.2 Notes

1. **Do not** hot-plug the power connector or motor cables to avoid damage.
 2. Recommended to pair with the store's **42-size magnetic-encoder stepper**.
 3. If you purchase the 42 stepper from the store together, the package will be **installed and calibrated** before shipment.
-

7 After-Sales & Tech Support

1. Bilibili: 张大头闭环伺服 — <https://space.bilibili.com/486903750>
2. Taobao shop: 张大头闭环伺服 — <https://shop113302005.taobao.com/>
3. CSDN blog: <https://blog.csdn.net/zhangdatou666>
4. Yuque online manual: <https://www.yuque.com/zhangdatouzhikong/gzng3d?#>
5. QQ (tech support): **424116966** (or @张小头 in the group)

6. QQ group: 262438510

8 Appendix

8.1 Checksum Methods

1. **Fixed 0x6B**: convenient for testing.
Example: 01 06 45 6B (calibration cmd).
2. **XOR**: XOR of all bytes before checksum.
Example: 01 06 45 XOR $\rightarrow$ XOR = $0x01 \wedge 0x06 \wedge 0x45$.
3. **CRC-8**:
Example: 01 06 45 CRC8 $\rightarrow$ CRC8 = cmd_calCRC8(p, 3), where p[] = {0x01, 0x06, 0x45}.

```
uint8_t cmd_calCRC8(uint8_t *p, uint8_t len) {  
    unsigned char i = 0, crc8 = p[0];  
    for(i = 1; i < len; i++) {  
        crc8 = crc8Table[crc8 ^ *(p + i)];  
    }  
    return crc8;  
}
```

```
const uint8_t crc8Table[256] = {  
    0x00, 0x5E, 0xBC, 0xE2, 0x61, 0x3F, 0xDD, 0x83, // 0  
    0xC2, 0x9C, 0x7E, 0x20, 0xA3, 0xFD, 0x1F, 0x41, // 8  
    0x9D, 0xC3, 0x21, 0x7F, 0xFC, 0xA2, 0x40, 0x1E, // 16  
    0x5F, 0x01, 0xE3, 0xBD, 0x3E, 0x60, 0x82, 0xDC, // 24  
    0x23, 0x7D, 0x9F, 0xC1, 0x42, 0x1C, 0xFE, 0xA0, // 32  
    0xE1, 0xBF, 0x5D, 0x03, 0x80, 0xDE, 0x3C, 0x62, // 40  
    0xBE, 0xE0, 0x02, 0x5C, 0xDF, 0x81, 0x63, 0x3D, // 48  
    0x7C, 0x22, 0xC0, 0x9E, 0x1D, 0x43, 0xA1, 0xFF, // 56  
    0x46, 0x18, 0xFA, 0xA4, 0x27, 0x79, 0x9B, 0xC5, // 64  
    0x84, 0xDA, 0x38, 0x66, 0xE5, 0xBB, 0x59, 0x07, // 72  
    0xDB, 0x85, 0x67, 0x39, 0xBA, 0xE4, 0x06, 0x58, // 80  
    0x19, 0x47, 0xA5, 0xFB, 0x78, 0x26, 0xC4, 0x9A, // 88  
    0x65, 0x3B, 0xD9, 0x87, 0x04, 0x5A, 0xB8, 0xE6, // 96
```

```
0xA7, 0xF9, 0x1B, 0x45, 0xC6, 0x98, 0x7A, 0x24, // 104
0xF8, 0xA6, 0x44, 0x1A, 0x99, 0xC7, 0x25, 0x7B, // 112
0x3A, 0x64, 0x86, 0xD8, 0x5B, 0x05, 0xE7, 0xB9, // 120
0x8C, 0xD2, 0x30, 0x6E, 0xED, 0xB3, 0x51, 0x0F, // 128
0x4E, 0x10, 0xF2, 0xAC, 0x2F, 0x71, 0x93, 0xCD, // 136
0x11, 0x4F, 0xAD, 0xF3, 0x70, 0x2E, 0xCC, 0x92, // 144
0xD3, 0x8D, 0x6F, 0x31, 0xB2, 0xEC, 0x0E, 0x50, // 152
0xAF, 0xF1, 0x13, 0x4D, 0xCE, 0x90, 0x72, 0x2C, // 160
0x6D, 0x33, 0xD1, 0x8F, 0x0C, 0x52, 0xB0, 0xEE, // 168
0x32, 0x6C, 0x8E, 0xD0, 0x53, 0x0D, 0xEF, 0xB1, // 176
0xF0, 0xAE, 0x4C, 0x12, 0x91, 0xCF, 0x2D, 0x73, // 184
0xCA, 0x94, 0x76, 0x28, 0xAB, 0xF5, 0x17, 0x49, // 192
0x08, 0x56, 0xB4, 0xEA, 0x69, 0x37, 0xD5, 0x8B, // 200
0x57, 0x09, 0xEB, 0xB5, 0x36, 0x68, 0x8A, 0xD4, // 208
0x95, 0xCB, 0x29, 0x77, 0xF4, 0xAA, 0x48, 0x16, // 216
0xE9, 0xB7, 0x55, 0x0B, 0x88, 0xD6, 0x34, 0x6A, // 224
0x2B, 0x75, 0x97, 0xC9, 0x4A, 0x14, 0xF6, 0xA8, // 232
0x74, 0x2A, 0xC8, 0x96, 0x15, 0x4B, 0xA9, 0xF7, // 240
0xB6, 0xE8, 0x0A, 0x54, 0xD7, 0x89, 0x6B, 0x35 // 248
};
```